

[Document Revision: 1.2]

Silicon Errata for the CYT2B9 Series Rev. B

This document describes the errata for CYT2B9 Rev. B Series of devices. Details include trigger conditions, scope of impact, available workarounds, and applicable silicon revisions. Contact your local Cypress Sales representative for further questions.

Part Numbers Affected

Part Number
CYT2B93CABES
CYT2B94CABES
CYT2B95CABES
CYT2B97CABES
CYT2B98CABES

CYT2B9 Rev. B Qualification Status

Product Status: Engineering Samples

CYT2B9 Rev. B Errata Summary

This table defines the errata applicable to CYT2B9 Rev. B Series of devices.

Items	Part Number	Silicon Revision	Fix Status
Errata ID 67 ConfigureFmInterrupt API assumes a parameter with 8 bytes boundary, but actual boundary is 4 bytes	CYT2B93CABES CYT2B94CABES CYT2B95CABES CYT2B97CABES CYT2B98CABES	Rev. B	No silicon fix planned. Use workaround.
Errata ID 68 SMPU/MPU/PPU protection region size is limited to 2 GB			No silicon fix planned. Use workaround.
Errata ID 69 DirectExecute API may return error if called with arguments placed in SRAM			No silicon fix planned. Use workaround.
Errata ID 96 CAN FD RX FIFO top pointer feature does not function as expected			No silicon fix planned. Use workaround.
Errata ID 97 CAN FD debug message handling state machine not get reset to Idle state when CANFD_CH_CCCR.INIT is set			No silicon fix planned. Use workaround.
Errata ID 98 TPIU peripheral ID mismatch			No silicon fix planned.
Errata ID 99 Unpredictable behavior of SROM API's if SRAM is not retained			Will be fixed in Rev. C

- 67. ConfigureFmInterrupt API assumes a parameter with 8 bytes boundary, but actual boundary is 4 bytes**
- **Problem Definition**
STATUS_ADDR_PROTECTED will be returned if ConfigureFmInterrupt API is called with arguments stored in SRAM with 4 bytes boundary of available SRAM or protected boundary SRAM.
 - **Parameters Affected**
N/A
 - **Trigger Condition(s)**
Call ConfigureFmInterrupt API with arguments stored in SRAM at 4 bytes boundary of available SRAM or protected boundary of SRAM.
 - **Scope of Impact**
ConfigureFmInterrupt API will fail by returning STATUS_ADDR_PROTECTED error status when called with argument having 4 bytes boundary of available SRAM or protected boundary of SRAM.
 - **Workaround**
Allow 4 bytes margin (i.e. assume that API parameter size as 8 and store the arguments) for API (ConfigureFmInterrupt) parameter.
 - **Fix Status**
No silicon fix planned. Use workaround.
- 68. SMPU/MPU/PPU protection region size is limited to 2 GB**
- **Problem Definition**
If SMPU/MPU/PPU protection block size is configured for 4 GB (PROT_SMPU_SMPU_STRUCT_ATT0.REGION.SIZE = 31), then during protection check in SROM, the value of the internal uint32 variable will overflow (4G = 0x1 0000 0000). Therefore, SROM assumes the protection size equals zero, and no protection will be applied.
 - **Parameters Affected**
N/A
 - **Trigger Condition(s)**
Configure SMPU/MPU/PPU to protect with region size equal to 4 GB or the region size with value 31u.
 - **Scope of Impact**
If SMPU/MPU/PPU is configured to protect region size of 4 GB, then SROM software does not apply any protection as per the request.
 - **Workaround**
Use two protection blocks of region size equal to 2 GB if 4 GB region size protection is required.
 - **Fix Status**
No silicon fix planned. Use workaround.
- 69. DirectExecute API may return error if called with arguments placed in SRAM memory**
- **Problem Definition**
If DirectExecute API is called in master PC (other than PC0 or PC1) with arguments in SRAM_SCRATCH_ADDR, then the API will return STATUS_ADDR_PROTECTED status.
 - **Parameters Affected**
N/A
 - **Trigger Condition(s)**
Call DirectExecute API with arguments in SRAM_SCRATCH_ADDR and master PC configured > 1.
 - **Scope of Impact**
DirectExecute API, if called with master PC configured > 1 and arguments in SRAM_SCRATCH_ADDR, the API will return STATUS_ADDR_PROTECTED.
 - **Workaround**
Call DirectExecute API with master PC0 or PC1, if arguments are stored in SRAM memory.
 - **Fix Status**
No silicon fix planned. Use workaround.

96. CAN FD RX FIFO top pointer feature does not function as expected

- **Problem Definition**
 RX FIFO top pointer function calculates the address for received messages in Message RAM by hardware. This address should be re-start back from the start address after reading all messages of RX FIFO n size (n: 0 or 1). However, the address does not re-start back from the start address when RX FIFO n size is set to 1 (CANFD_CH_RXFnC.FnS = 0x01). This results in CPU/DMA to read messages from the wrong address in Message RAM.
- **Parameters Affected**
 N/A
- **Trigger Condition(s)**
 RX FIFO top pointer function is used when RX FIFO n size set to 1 element (CANFD_CH_RXFnC.FnS = 0x01).
- **Scope of Impact**
 Received message cannot be correctly read by using RX FIFO top pointer function, when RX FIFO n size set to 1 element.
- **Workaround**
 Any of the following.
 1) Set RX FIFO n size to 2 or more when using RX FIFO top pointer function.
 2) Do not use RX FIFO top pointer function when RX FIFO n size set to 1 element. Instead of RX FIFO top pointer, read received messages from the Message RAM directly.
- **Fix Status**
 No silicon fix planned. Use workaround.

97. CAN FD debug message handling state machine not get reset to Idle state when CANFD_CH_CCCR.INIT is set

- **Problem Definition**
 If either CANFD_CH_CCCR.INIT bit is set by the Host or when the M_TTCAN module enters BusOff state, the debug message handling state machine stays in its current state instead of being reset to Idle state. Configuring the bit CANFD_CH_CCCR.CCE does not change CANFD_CH_RXF1S.DMS.
- **Parameters Affected**
 N/A
- **Trigger Condition(s)**
 Either CANFD_CH_CCCR.INIT bit is set by the Host or when the M_TTCAN module enters BusOff state.
- **Scope of Impact**
 The errata is limited to the use case when the Debug on CAN functionality is active. Normal operation of CAN module is not affected, in which case the debug message handling state machine always remains in Idle state. In the described use case, the debug message handling state machine is stopped and remains in the current state signaled by the bit CANFD_CH_RXF1S.DMS. In case CANFD_CH_RXF1S.DMS is set to 0b11, DMA request remains active.
- **Workaround**
 In case the debug message handling state machine has stopped while CANFD_CH_RXF1S.DMS is 0b01 or 0b10, it can be reset to Idle state by hardware reset or by reception of debug messages after CANFD_CH_CCCR.INIT is reset to zero.
- **Fix Status**
 No silicon fix planned. Use workaround.

98. TPIU peripheral ID mismatch

- **Problem Definition**
 TPIU peripheral ID indicates that it is M3-TPIU instead of M4-TPIU.
- **Parameters Affected**
 N/A
- **Trigger Condition(s)**
 When debugger reads PID registers for component identification.

- **Scope of Impact**
The debuggers read the TPIU as M3-TPIU and no other impact other than this.
- **Workaround**
No specific workaround required. Debuggers can use trace features.
- **Fix Status**
No silicon fix planned.

99. Unpredictable behavior of SROM API's if SRAM is not retained

- **Problem Definition**
Any of SROM API's assumes the contents in the last 2 KB location of SRAM1 is retained at all the times. If any of SROM API's are called after SRAM1 is not retained in all Active, LP Active, Sleep, LP Sleep, DeepSleep modes, its behavior is unpredictable.
- **Parameters Affected**
N/A
- **Trigger Condition(s)**
The following conditions are met:
(1) SRAM1 is not retained in all Active, LP Active, Sleep, LP Sleep, DeepSleep modes.
(2) Any of SROM API's are called after (1).
- **Scope of Impact**
Unpredictable behavior of SROM API's if SRAM1 is not retained in all Active, LP Active, Sleep, LP Sleep, DeepSleep modes.
- **Workaround**
Any of the following:
(A) Retain SRAM1 in all Active, LP Active, Sleep, LP Sleep, DeepSleep modes.
(B) Assert high voltage reset (POR, BOD, OVD, OCD, WDT, Hibernate wakeup, or XRES_L) or software reset through the AIRCR.SYSRESETREQ register of each CPU if any of SROM API's are called after SRAM1 is not retained in all Active, LP Active, Sleep, LP Sleep, DeepSleep modes.
- **Fix Status**
Will be fixed in silicon Rev. C.
The revised device's SROM will be updated to move system call relevant contents from last 2 KB of SRAM1 to first 2 KB of SRAM0. Therefore, user needs to move application execution from SRAM0 zero offset to SRAM0 2 KB offset.

Document History Page

Document Title: Silicon Errata for the CYT2B9 Series Rev. B		
Issue Date	Document Rev.	Description of Change
July 11, 2019	1.0	Initial release
Aug 14, 2019	1.1	Modified the description in errata ID 67 to 69. Added errata ID 96 to 98.
Oct 9, 2019	1.2	Added errata ID 99.



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